WoSM Academy

M201-F27 · Calculus II

# Power series

Radius of convergence; differentiation and integration term by term

Dr. Ali Khudiyev
WoSM Academy

Lecture 05 · Unit 5 · Fall 2027


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# Last time

- $f_n \to f$ **uniformly** on $D$ if $\sup_{x \in D} |f_n(x) - f(x)| \to 0$.
- Uniform limits of continuous functions are continuous; on $[a, b]$ we may swap lim and $\int$; derivatives need uniform convergence of $f_n'$.
- **M-test**: $|f_k(x)| \le M_k$ with $\sum M_k < \infty$ gives uniform convergence of $\sum f_k$.
- $\sum x^k = \frac{1}{1-x}$ converges uniformly on every $[-a, a]$ with $a < 1$, so we could integrate and differentiate it term by term.

Today: the same for every power series.

# This lecture

01 Power series

02 The radius of convergence

03 Calculus with power series

04 New series from old

05 Summary and practice

Reading: G. Strang, E. Herman, *Calculus, Volume 2* (OpenStax), Sections 6.1 and 6.2.

01

# Power series

# Power series

**Definition**

A **power series centred at** $a$ is a series

$$\sum_{k=0}^{\infty} c_k (x-a)^k = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots$$

with **coefficients** $c_k \in \mathbb{R}$. (We read $(x-a)^0$ as 1, also at $x = a$.)

- A polynomial "of infinite degree". Its partial sums are polynomials.
- It always converges at $x = a$, to $c_0$.
- For which $x$ does it converge? What function does it define? Can we differentiate and integrate it like a polynomial?

# Four examples

With the ratio test (Lecture 3), for $x \neq 0$:

- $\sum x^k$:
- $\sum \dfrac{x^k}{k!}$:
- $\sum k!\, x^k$:
- $\displaystyle\sum_{k=1}^{\infty} \frac{x^k}{k}$:

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- $\sum x^k$: ratio $|x|$. Converges for $|x| < 1$ only.
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- $\sum k!\, x^k$: ratio $(k+1)|x| \to \infty$. Converges only at $x = 0$.
- $\displaystyle\sum_{k=1}^{\infty} \frac{x^k}{k}$: ratio $\frac{k}{k+1}|x| \to |x|$. Converges for $|x| < 1$ and diverges for $|x| > 1$. At $x = 1$: the harmonic series, divergent. At $x = -1$: the alternating harmonic series, convergent. So: $[-1, 1)$.

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In each case the set of convergence is an interval centred at $0$ — possibly a single point or all of $\mathbb{R}$.

02

# The radius of convergence

# Convergence spreads inwards

**Lemma (centre $a = 0$; otherwise replace $x$ by $x - a$)**

If $\sum c_k x_0^k$ converges for some $x_0 \neq 0$, then $\sum c_k x^k$ converges absolutely for every $|x| < |x_0|$, and uniformly on $[-r, r]$ for every $r < |x_0|$.

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**Proof.**

The terms of a convergent series tend to 0, so they are bounded: $|c_k x_0^k| \leq M$ for all $k$. For $|x| \leq r$, $|c_k x^k| = |c_k x_0^k| \left|\frac{x}{x_0}\right|^k \leq M q^k$ with $q = \frac{r}{|x_0|} < 1$. As $\sum M q^k$ converges, the M-test applies. □

# The radius of convergence

**Theorem**

For every power series $\sum c_k (x-a)^k$ there is an $R$ with $0 \le R \le \infty$, the **radius of convergence**, such that the series

- converges absolutely if $|x-a| < R$, uniformly on every $[a-r, a+r]$ with $r < R$;
- diverges if $|x-a| > R$.

Proof: let $R$ be the supremum of $|x-a|$ over all $x$ where the series converges, and apply the lemma.

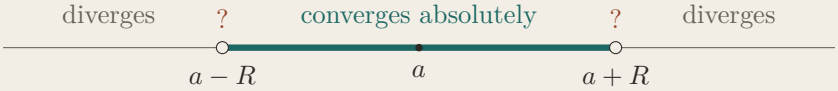
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# Computing $R$

**Theorem**

If $c_k \neq 0$ and $\left|\dfrac{c_{k+1}}{c_k}\right| \to L$, then $R = \dfrac{1}{L}$ (with $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$). The same holds if $\sqrt[k]{|c_k|} \to L$.

Proof: the ratio of consecutive terms is $\left|\dfrac{c_{k+1}(x-a)^{k+1}}{c_k(x-a)^k}\right| \to L\,|x-a|$, which is below 1 if and only if $|x-a| < \frac{1}{L}$. Apply the ratio test (Lecture 3).

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**Remark**

The **Cauchy–Hadamard formula** $R = 1/\limsup_k \sqrt[k]{|c_k|}$ always holds, even when the limits above do not exist. We will not need it.

# Examples

- $\displaystyle\sum_{k=1}^{\infty} \frac{(x-2)^k}{k\,3^k}$
- $\displaystyle\sum_{k=0}^{\infty} \frac{x^{2k}}{4^k}$
- $\displaystyle\sum_{k=1}^{\infty} k^k x^k$

# Examples

- $\displaystyle\sum_{k=1}^{\infty} \frac{(x-2)^k}{k\,3^k}$
  $\left|\frac{c_{k+1}}{c_k}\right| = \frac{k}{3(k+1)} \to \frac{1}{3}$, so $R = 3$: converges for $-1 < x < 5$. At $x = 5$: $\sum \frac{1}{k}$, divergent; at $x = -1$: $\sum \frac{(-1)^k}{k}$, convergent. Interval: $[-1, 5)$.
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- $\displaystyle\sum_{k=0}^{\infty} \frac{x^{2k}}{4^k}$
  Every other coefficient is 0, so the formula does not apply. Ratio test on the terms: $\frac{x^2}{4} < 1$ iff $|x| < 2$, so $R = 2$. (Sum: $\frac{1}{1 - x^2/4}$.)
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- $\displaystyle\sum_{k=1}^{\infty} k^k x^k$
  $\sqrt[k]{|c_k|} = k \to \infty$, so $R = 0$: it converges only at $x = 0$.

# At the endpoints anything can happen

| **Series** | $R$ | **At** $x = 1$ | **At** $x = -1$ |
|---|---|---|---|
| $\sum x^k$ | 1 | diverges | diverges |
| $\sum \frac{x^k}{k}$ | 1 | diverges | converges conditionally |
| $\sum \frac{(-1)^k x^k}{k}$ | 1 | converges conditionally | diverges |
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Check the endpoints separately, with the tests of Lectures 2 and 3.

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> **Abel's theorem, without proof**
>
> If the series converges at $x = a + R$, its sum is continuous there from the left: $\lim_{x \to (a+R)^-} \sum c_k (x-a)^k = \sum c_k R^k$. Likewise at $a - R$.

03

# Calculus with power series

# Differentiation and integration term by term

**Theorem**

Let $\sum c_k (x-a)^k$ have radius $R > 0$ and sum $f(x)$ for $|x-a| < R$. Then $f$ is continuous and differentiable there, and

$$f'(x) = \sum_{k=1}^{\infty} k\, c_k (x-a)^{k-1}, \qquad \int_a^x f(t)\,\mathrm{d}t = \sum_{k=0}^{\infty} \frac{c_k}{k+1}(x-a)^{k+1}.$$

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Both new series have the same radius $R$.

- Inside the interval of convergence a power series behaves like a polynomial.
- The radius stays the same; the behaviour at the endpoints may change.

# Proof

**Proof.**

Take $a = 0$ and $|x| < r < R$. The series converges at $r$, so $|c_k r^k| \le M$. Then

$$|k c_k x^{k-1}| \le \frac{M}{r}\, k\, q^{k-1}, \qquad q = \frac{|x|}{r} < 1,$$

and $\sum k q^{k-1}$ converges (ratio test). So the derivative series converges absolutely for $|x| < R$, and by the M-test uniformly on every $[-r', r']$ with $r' < R$. Lecture 4: we may differentiate term by term there. For integrals: on $[0, x]$ the series converges uniformly; swap $\int$ and $\sum$. The radius cannot grow: $|c_k x^k| \le |x| \cdot |k c_k x^{k-1}|$ for $k \ge 1$. □

# Coefficients are derivatives

**Corollary**

If $f(x) = \sum c_k (x-a)^k$ for $|x-a| < R$ with $R > 0$, then $f$ has derivatives of every order there, and

$$c_k = \frac{f^{(k)}(a)}{k!} \qquad (k = 0, 1, 2, \dots).$$

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- **Uniqueness**: two power series centred at $a$ that agree near $a$ have the same coefficients. However we find a power series for $f$, it is the same one.
- Next lecture: start from $f$ and use $c_k = \frac{f^{(k)}(a)}{k!}$ — the Taylor series.

# Endpoints can change

**Example**

- $\sum_{k=1}^{\infty} \frac{x^k}{k}$ converges on $[-1, 1)$. Its derivative $\sum_{k=1}^{\infty} x^{k-1}$ converges on $(-1, 1)$ only.
- $\sum_{k=1}^{\infty} \frac{x^k}{k^2}$ converges on $[-1, 1]$. Its derivative $\sum_{k=1}^{\infty} \frac{x^{k-1}}{k}$ converges on $[-1, 1)$.

Differentiating can lose an endpoint; integrating can gain one. The theorem is about the open interval $(a - R, a + R)$.

04

# New series from old

# Substitution into the geometric series

Start from $\frac{1}{1-u} = \sum_{k=0}^{\infty} u^k$ for $|u| < 1$:

- $u = -x$: $\dfrac{1}{1+x} = \sum_{k=0}^{\infty} (-1)^k x^k, \quad |x| < 1.$
- $u = -x^2$: $\dfrac{1}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k x^{2k}, \quad |x| < 1.$
- $u = \frac{x}{2}$: $\dfrac{1}{2-x} = \dfrac{1}{2} \cdot \dfrac{1}{1-x/2} = \sum_{k=0}^{\infty} \dfrac{x^k}{2^{k+1}}, \quad |x| < 2.$
- Another centre, $u = -(x-1)$: $\dfrac{1}{x} = \dfrac{1}{1+(x-1)} = \sum_{k=0}^{\infty} (-1)^k (x-1)^k,$ $|x-1| < 1.$

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Note $\frac{1}{1+x^2}$ is defined for every $x$, yet its series has $R = 1$. Something at $x = \pm i$ in the complex plane decides that — a story for complex analysis.

# Integrating term by term

- $\ln(1+x) = \displaystyle\int_0^x \frac{\mathrm{d}t}{1+t} = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \qquad (|x| < 1)$
- $-\ln(1-x) = \displaystyle\sum_{k=1}^{\infty} \frac{x^k}{k} \qquad (|x| < 1)$
- $\arctan x = \displaystyle\int_0^x \frac{\mathrm{d}t}{1+t^2} = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{2k+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots \qquad (|x| < 1)$

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**Remark**

At $x = 1$ the series for $\ln(1+x)$ and $\arctan x$ converge (alternating series). By Abel's theorem their sums are $\ln 2$ and $\arctan 1 = \frac{\pi}{4}$. Next lecture proves both directly, with an estimate of the error.

# Differentiating term by term

From $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$, for $|x| < 1$:

$$\frac{1}{(1-x)^2} = \sum_{k=1}^{\infty} k x^{k-1}, \qquad \frac{x}{(1-x)^2} = \sum_{k=1}^{\infty} k x^k, \qquad \frac{x(1+x)}{(1-x)^3} = \sum_{k=1}^{\infty} k^2 x^k.$$

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**Example (A binary counter)**

An increment flips the trailing 1s and one 0: `1011` $\to$ `1100` flips 3 bits. A fraction $\frac{1}{2^k}$ of all values end in `0` and then exactly $k-1$ ones, so the average number of bit flips per increment is $\sum_{k=1}^{\infty} \frac{k}{2^k} = \frac{1/2}{(1-1/2)^2} = 2$.

# The exponential function

Let $E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$. Its radius is $\infty$, and term by term

$$E'(x) = \sum_{k=1}^{\infty} \frac{k\,x^{k-1}}{k!} = \sum_{k=1}^{\infty} \frac{x^{k-1}}{(k-1)!} = E(x), \qquad E(0) = 1.$$

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**Theorem**

$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ for every $x \in \mathbb{R}$.

Proof: $\big(E(x)e^{-x}\big)' = E(x)e^{-x} - E(x)e^{-x} = 0$, so $E(x)e^{-x} = E(0)e^0 = 1$.

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- $x = 1$: $e = \sum_{k=0}^{\infty} \frac{1}{k!}$, so $\sum_{k=1}^{\infty} \frac{1}{k!} = e - 1$ (Lecture 2).
- $x = -1$: $\frac{1}{e} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}$, so $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k!} = 1 - \frac{1}{e}$ (Lecture 3).

# Multiplying power series

**Theorem (Cauchy product)**

If $\sum a_k x^k$ and $\sum b_k x^k$ both converge for $|x| < R$, then for $|x| < R$

$$\Big(\sum_{k=0}^{\infty} a_k x^k\Big)\Big(\sum_{k=0}^{\infty} b_k x^k\Big) = \sum_{k=0}^{\infty} c_k x^k, \qquad c_k = \sum_{j=0}^{k} a_j b_{k-j}.$$

Both series converge absolutely, so all products $a_j x^j \cdot b_m x^m$ may be added in any order (Lecture 3) — in particular grouped by $j + m = k$. We omit the details.

# Multiplying power series: an example

**Example**

$\dfrac{1}{(1-x)^2} = \left(\sum x^k\right)^2 = \sum_{k=0}^{\infty}(k+1)x^k$, since $c_k = \sum_{j=0}^{k} 1 \cdot 1 = k+1$. The same as by differentiation.

The sequence $(c_k)$ is the *convolution* of $(a_k)$ and $(b_k)$. In Unit 7 this turns counting problems into algebra.

05

# Summary and practice

# Summary

- A power series $\sum c_k(x-a)^k$ has a **radius of convergence** $R \in [0,\infty]$: absolute convergence for $|x-a| < R$, divergence for $|x-a| > R$. Check the endpoints separately.
- $R = 1/L$ if $\left|\frac{c_{k+1}}{c_k}\right| \to L$ or $\sqrt[k]{|c_k|} \to L$.
- Uniform convergence on every $[a-r, a+r]$ with $r < R$.
- Inside, differentiate and integrate term by term; the radius does not change.
- $c_k = \frac{f^{(k)}(a)}{k!}$: a function has at most one power series around $a$.
- New series from old: substitute, integrate, differentiate, multiply. Results so far: $\frac{1}{1-x}$, $\ln(1+x)$, $\arctan x$, $e^x$.

# Practice

1. Find the radius and the interval of convergence:

   (a) $\sum_{k=1}^{\infty} \frac{x^k}{k\,2^k}$ (b) $\sum_{k=1}^{\infty} k^2 x^k$ (c) $\sum_{k=1}^{\infty} \frac{(x+1)^k}{k^2}$ (d) $\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$

2. Find a power series for $\frac{x}{1+x^2}$ around $0$, and its radius.
3. Find a power series for $\ln(2+x)$ around $0$, and its radius.
4. Find $\sum_{k=1}^{\infty} \frac{k}{3^k}$.
5. Use $\arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}$ to write $\pi$ as a series. How fast does it converge?
6. Show that if $\sum c_k x^k$ has radius $R$, then $\sum c_k x^{2k}$ has radius $\sqrt{R}$.
7. Find $\left(\sum_{k=0}^{\infty} x^k\right)\left(\sum_{k=0}^{\infty} (-1)^k x^k\right)$ as a Cauchy product, and check your answer another way.

# Answers

1. (a) $R = 2$, interval $[-2, 2)$. (b) $R = 1$, interval $(-1, 1)$. (c) $R = 1$, interval $[-2, 0]$. (d) $R = \infty$ (ratio of terms $\frac{x^2}{(2k+1)(2k+2)} \to 0$).
2. $x \sum_{k=0}^{\infty} (-1)^k x^{2k} = \sum_{k=0}^{\infty} (-1)^k x^{2k+1}$, $R = 1$.
3. $\ln 2 + \ln\!\left(1 + \frac{x}{2}\right) = \ln 2 + \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k\, 2^k}$, $R = 2$.
4. $\frac{x}{(1-x)^2}$ at $x = \frac{1}{3}$: $\frac{1/3}{4/9} = \frac{3}{4}$.
5. $\pi = 2\sqrt{3} \sum_{k=0}^{\infty} \frac{(-1)^k}{3^k (2k+1)}$. The terms shrink by a factor of about 3 each time: roughly one decimal digit for every two terms.
6. $\sum c_k (x^2)^k$ converges absolutely if $x^2 < R$ and diverges if $x^2 > R$.
7. $c_k = \sum_{j=0}^{k} (-1)^{k-j}$ is 1 for even $k$ and 0 for odd $k$: the product is $\sum x^{2k} = \frac{1}{1-x^2} = \frac{1}{1-x} \cdot \frac{1}{1+x}$.

# Next time

**Lecture 06 — Taylor series** (Unit 6)

- Which functions are power series? Taylor and Maclaurin series
- Taylor's theorem and estimates of the remainder
- Series for sin, cos, $(1+x)^\alpha$; and, at last, $\ln 2$ and $\frac{\pi}{4}$

Reading: Strang & Herman, *Calculus, Volume 2*, Sections 6.3 and 6.4.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.